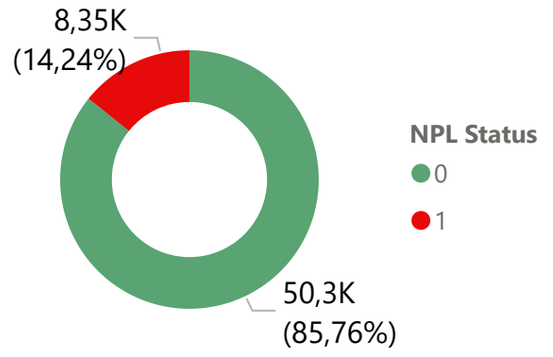
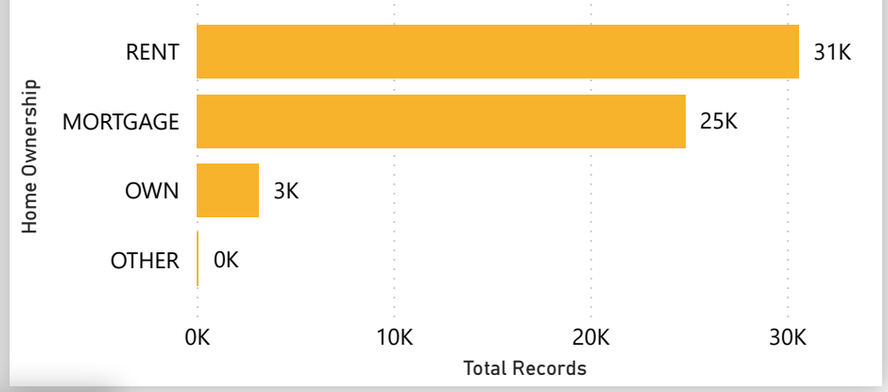
Data Overivew

59K	13	0	$540,6M	9,22K	10,68
Total Records	Total Features	Missing Values	Portfolio Value	Average Loan Amount	Average Interest Rate

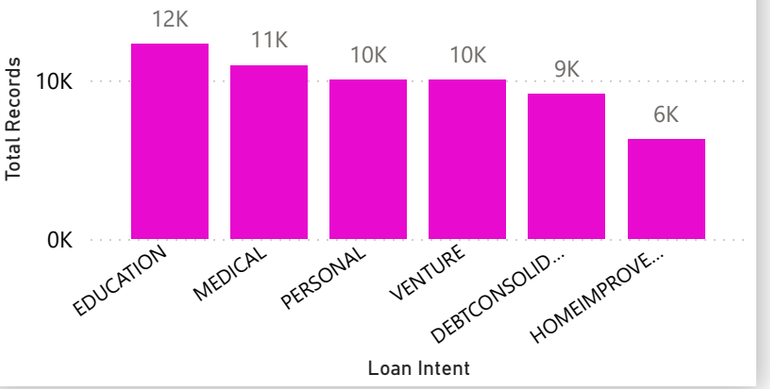
Loan Status Distribution



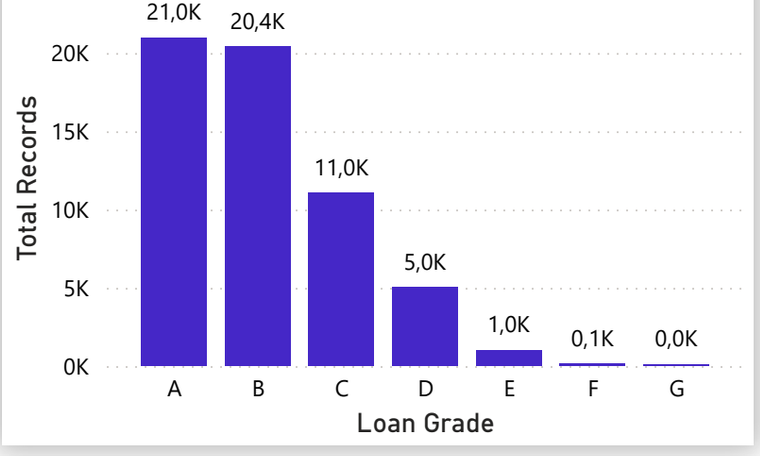
Home Ownership Distribution



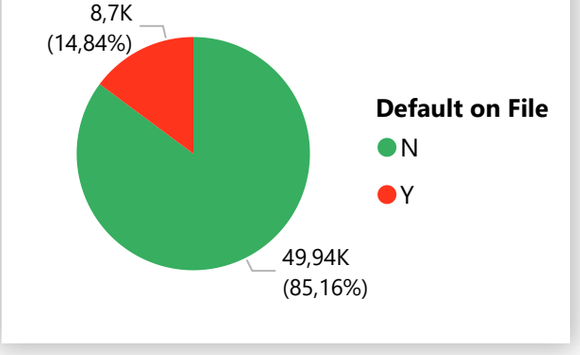
Loan Intent Types



Loan Grade Distribution



CB Default Distribution



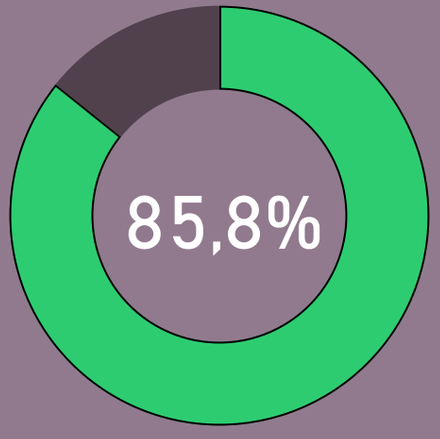
27,55	64,05K	4,70
Avg of Age	Avg of Income	Avg of Emp Length

0,17	5,81
Avg of DTI	Avg of Credit Hist

Loan Summary



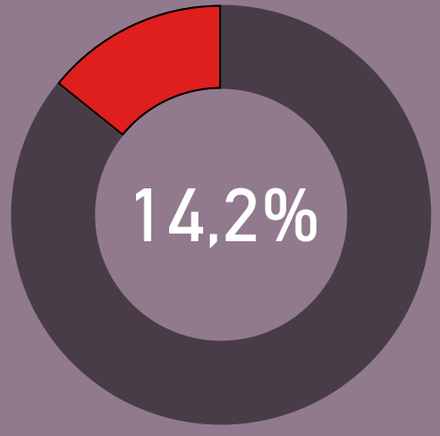
59K	$540,6M	$493,7M	10,68	0,17
Total Loan Applications	Total Amount Funded	Total Amount Received	Average Interest Rate	Average DTI



Good Loan Applications
50K

Good Loan Funded Amount
$447,1M

Good Loan Total Received
$493,7M



Bad Loan Applications
8350

Bad Loan Funded Amount
$93,5M

Bad Loan Total Received
0

CB Default
All

Loan Grade
All

Intent Type
All

Home Status
All

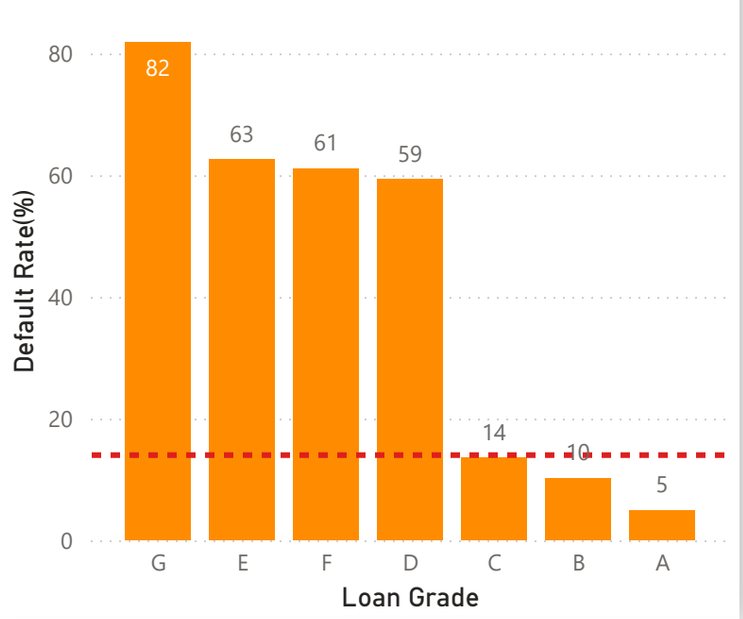
$46,6M	-$46,9M	-$93,5M
Good Loan Profit	Net Portfolio Result	Bad Loan Loss

Loss Given Default is accepted as 0

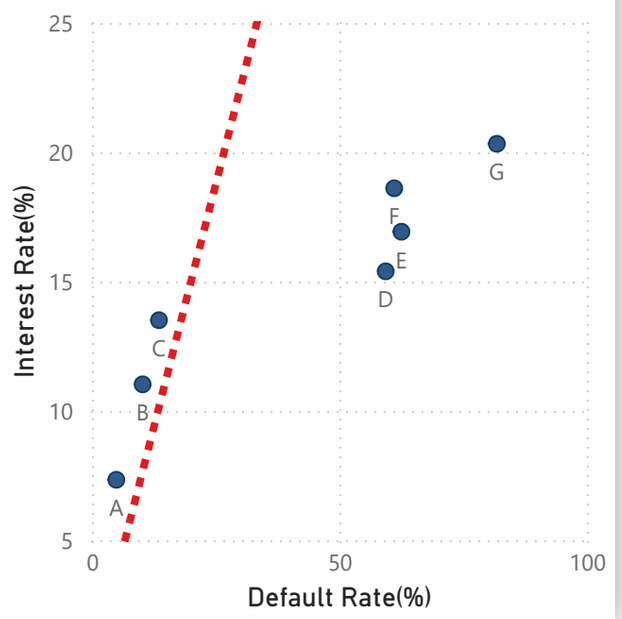
LGD

Risk Analysis Dashboard

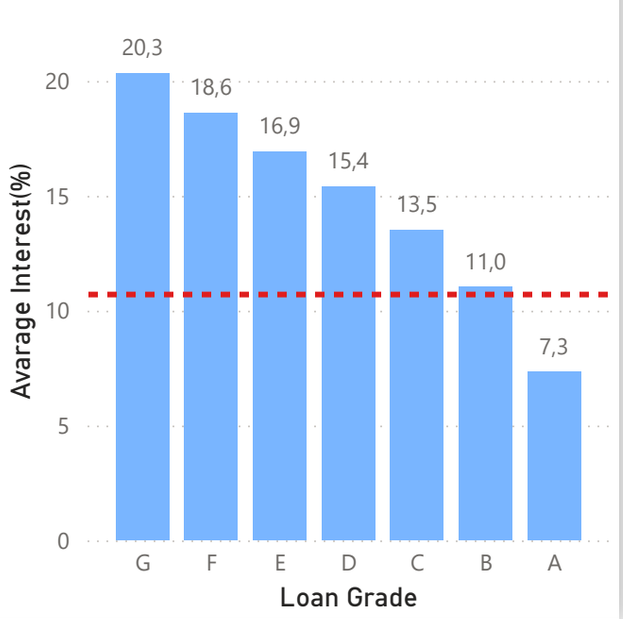
Default Rates by Grade



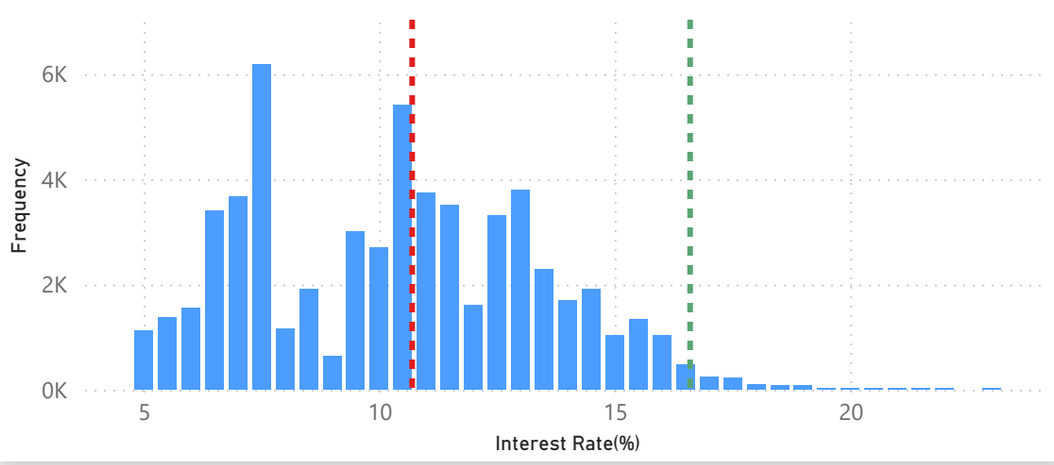
Risk-Return Matrixi



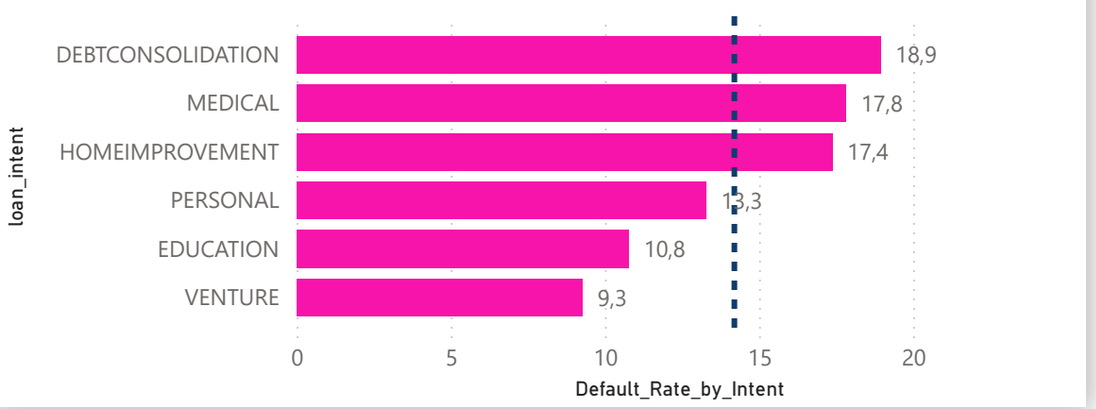
Avarage Interest by Loan Grade



Interest Rate Distribution



Default Rate (%) by Loan Intent



Cross-Segment Risk Assessment

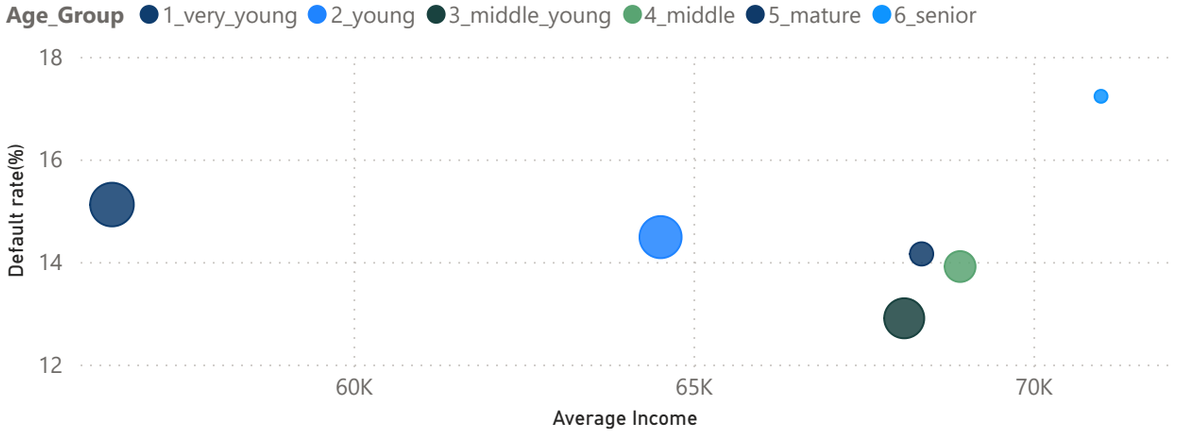
Default Rates by Loan Grade X Loan Intent

loan_grade	DEBTCONSOLIDATION	EDUCATION	HOMEIMPROVEMENT	MEDICAL	PERSONAL	VENTURE	Total
A	5,44	4,67	5,79	6,15	5,12	2,79	**4,92**
B	11,56	9,47	13,35	10,67	11,97	5,99	**10,23**
C	15,61	10,77	19,75	14,71	12,97	10,47	**13,54**
D	93,81	36,42	51,73	87,05	44,97	40,58	**59,36**
E	97,48	51,40	50,58	93,68	45,14	48,39	**62,54**
F	95,24	39,29	54,55	96,55	45,83	36,00	**61,07**
G	100,00	100,00	0,00	100,00	57,14	75,00	**81,82**
Total	**18,93**	**10,77**	**17,37**	**17,83**	**13,28**	**9,28**	**14,24**

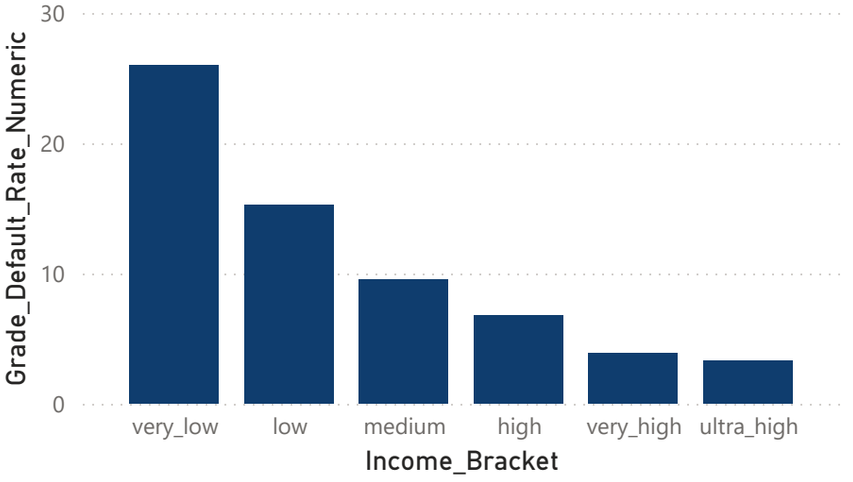
Home Ownership Status

- [ ] MORTGAGE
- [ ] OTHER
- [ ] OWN
- [ ] RENT

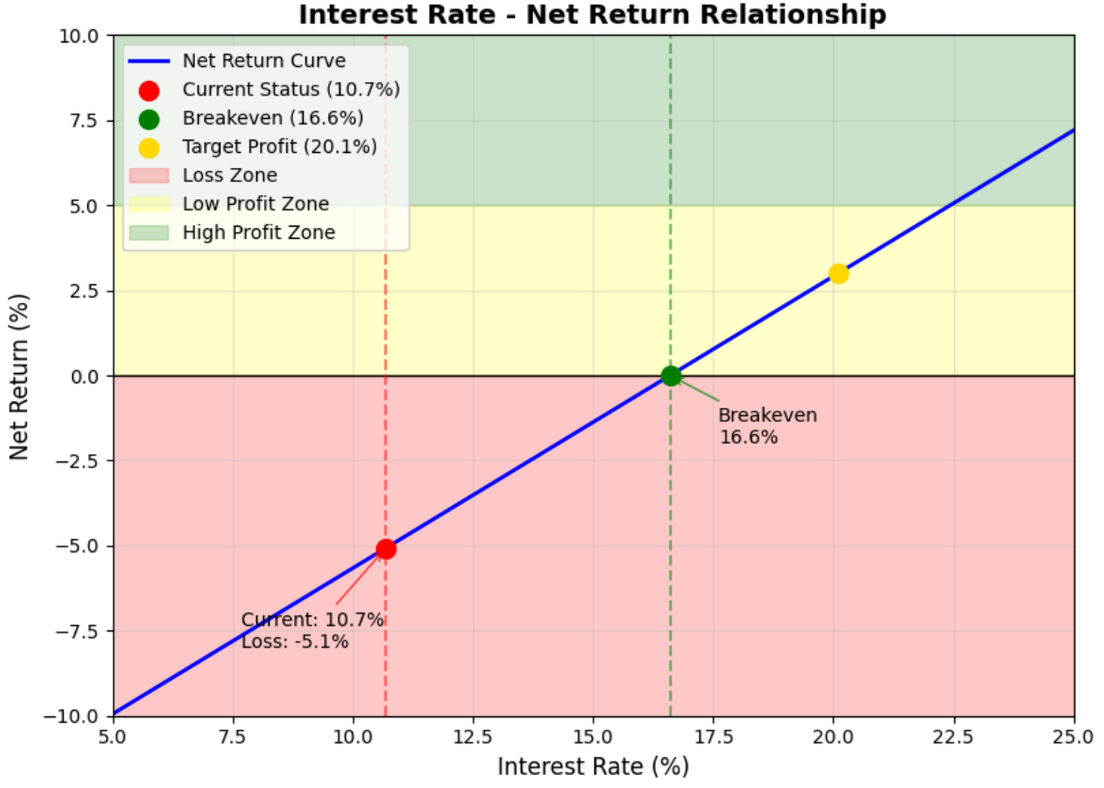
Income-Risk Relationship Across Age Segments

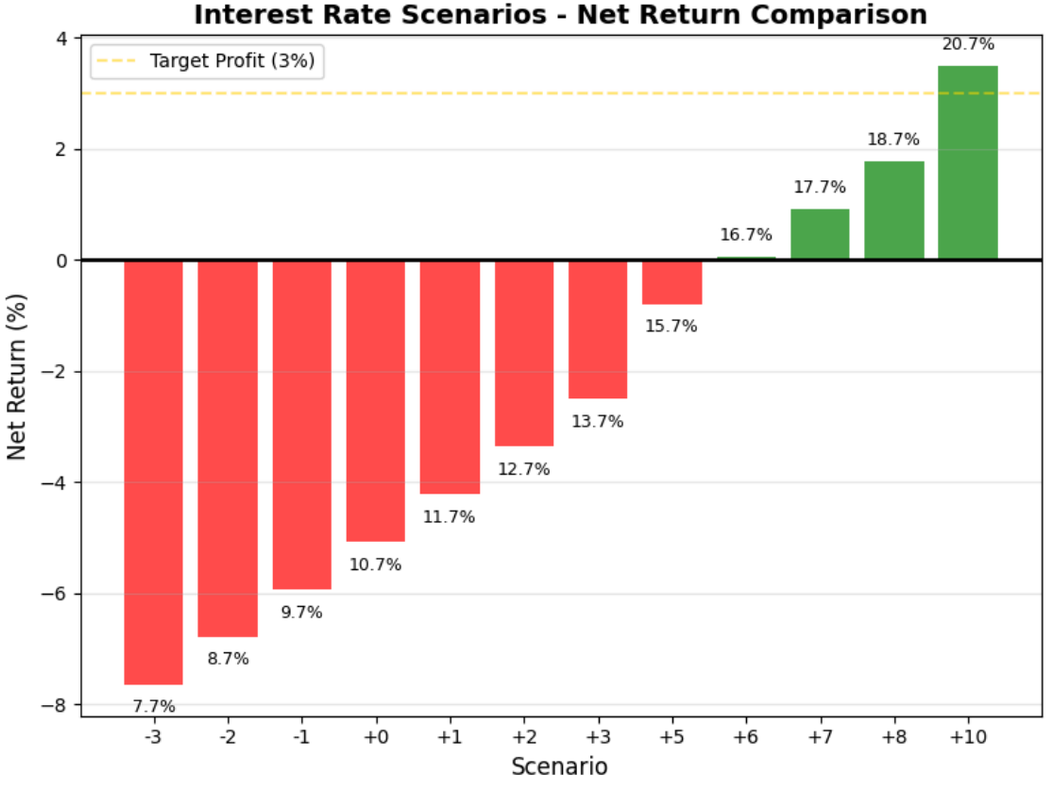


Grade_Default_Rate_Numeric by Income_Bracket



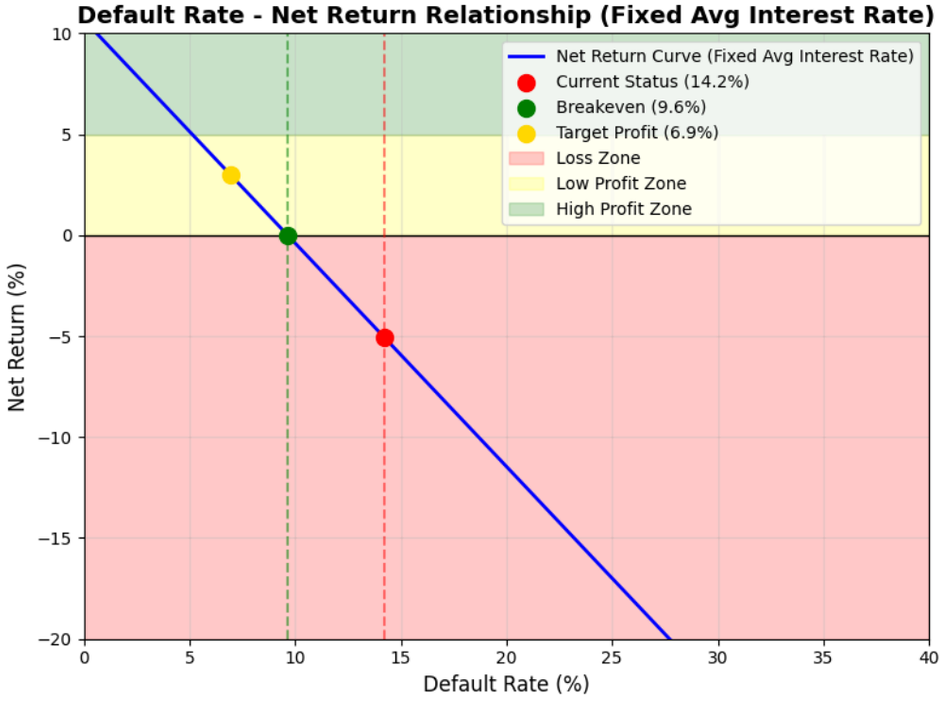
Cross-Segment Risk Assessment

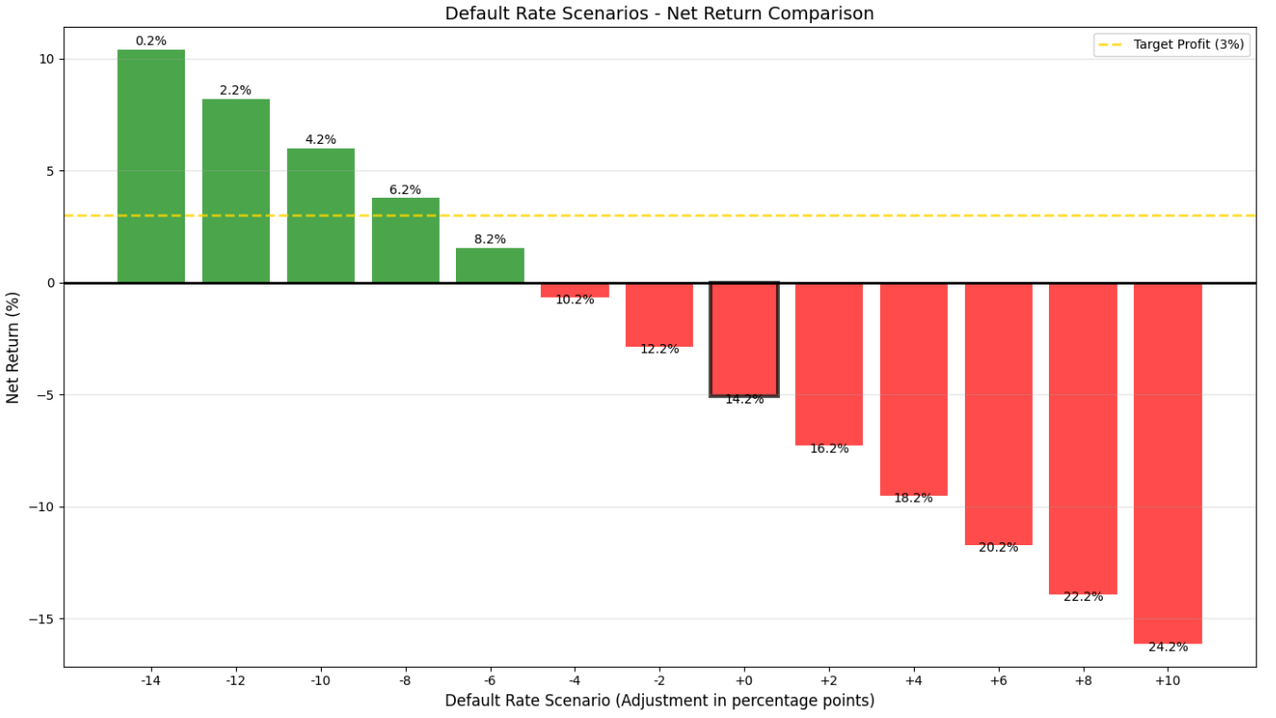




loan_grade	Grade_Count	Grade_Default_Rate	Grade_Avg_Interest	Grade_Avg_Loan	Grade_Net_Return%	Grade_Avg_Net_Amount	Grade_Status
A	20984	4,92%	7,34	8.447,00	2,06	$174	Profitable
B	20400	10,23%	11,03	9.622,00	-0,32	-$31	Loss-Making
C	11036	13,54%	13,51	9.010,00	-1,86	-$168	Loss-Making
D	5034	59,36%	15,40	10.521,00	-53,10	-$5.587	Loss-Making
F	149	61,07%	18,61	12.834,00	-53,83	-$6.909	Loss-Making
E	1009	62,54%	16,93	12.176,00	-56,19	-$6.842	Loss-Making
G	33	81,82%	20,33	12.559,00	-78,12	-$9.811	Loss-Making
Total	**58645**	**14,24%**	**10,68**	**9.218,00**	**-5,08**	**-$468**	**Loss-Making**

Default Rate - Net Return Relationship





📊 NET RETURN CALCULATION FORMULA

Net Return = Interest Rate × (1 - Default Rate) - Default Rate
- LGD (Loss Given Default) = 100%
- Assumes total principal loss on default
- Fixed average interest rate scenario

PORTFOLIO INSIGHTS SUMMARY

Current Status: 14.2% default rate = -5.1% net return
Breakeven Point: 9.6% default rate = 0% net return
Target Profit: 6.9% default rate = 3% net return
Required Improvement: -7.3 percentage points default reduction

Model

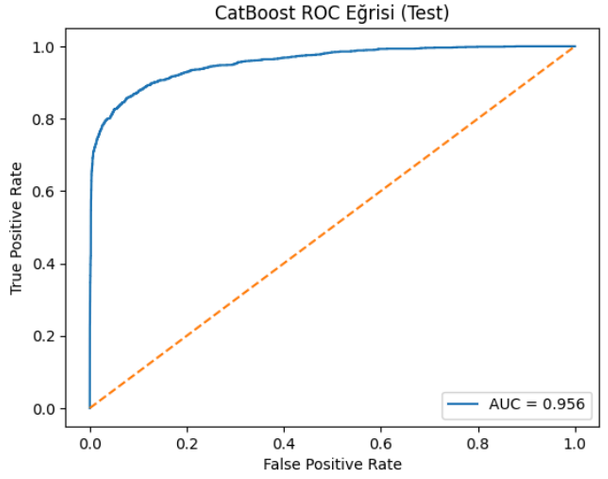
95.6%
ROC-AUC:

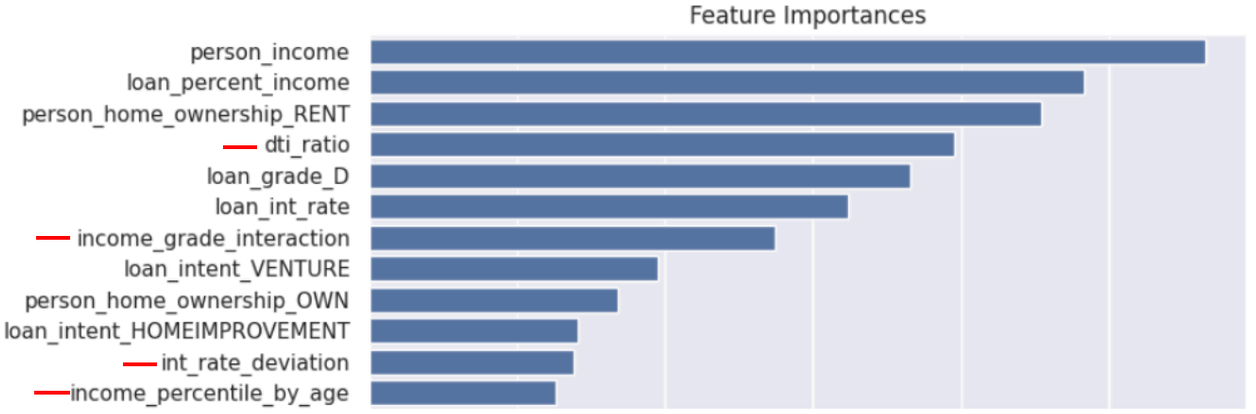
92.6%
Accuracy

70.5%
Precision

83.0%
Recall

76.3%
F1-Score





Confusion Matrix

Metric	Sum of Predicted_0	Sum of Predicted_1
Actual_0	9479	580
Actual_1	283	1387
Total	**9762**	**1967**

MODEL SPECIFICATIONS

Algorithm: CatBoost Classifier
Validation: 10-fold Stratified Cross-Validation
Hyperparameter Optimization: RandomizedSearchCV (10 iterations)
Class Balancing: Auto-weighted for imbalanced data
Categorical Features: Native handling (no encoding needed)
Training Size: 46,916 observations
Test Size: 11,729 observations